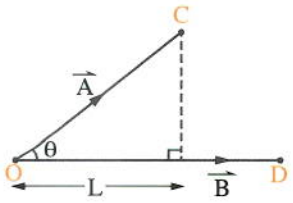
Unit 3

Lesson Three

Scalar product of two vectors

• The algebraic component (projection) of a vector in direction of another vector :

Let the two non-zero vectors $\vec{A}$ and $\vec{B}$ in 3 D - space are represented geometrically by two directed segments $\overrightarrow{OC}$ and $\overrightarrow{OD}$ drawn from the same starting point O



- θ is the measure of the smaller angle between the two vectors $\vec{A}$ and $\vec{B}$
- $\| \vec{A} \|$ is the length of the directed segment $\overrightarrow{OC}$
- $\| \vec{B} \|$ is the length of the directed segment $\overrightarrow{OD}$
- L is the length of the projection (component) of vector $\vec{A}$ in direction of $\vec{B}$

From the geometry of the figure : $\therefore L = OC \cos\theta$

$\therefore$ The component of vector $\vec{A}$ in direction of vector $\vec{B} = \| \vec{A} \| \cos\theta$

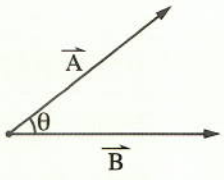
• The scalar product of two vectors (Dot product) :

Definition

If $\vec{A}$, $\vec{B}$ are two vectors including an angle of measure θ then the product of

norm of one of them and

the component of the other vector on it is known as the scalar product of these vectors and denoted as $\vec{A} \cdot \vec{B}$



i.e. $\vec{A} \cdot \vec{B} = \| \vec{A} \| \| \vec{B} \| \cos\theta$

i.e. The scalar product of two vectors is the scalar quantity equal to the product of the norm of one vector by the norm of the 2nd vector by cosine of the smaller angle between them.

For example :

If $\| \vec{A} \| = 5$, $\| \vec{B} \| = 4$, $\theta = 60°$

then $\vec{A} \cdot \vec{B} = \| \vec{A} \| \| \vec{B} \| \cos\theta = 5 \times 4 \times \cos 60° = 10$

Remarks

1. In the definition of the scalar product , the smaller angle is used , but if the greater angle is used instead , it gives the same answer because $\cos(360° - \theta) = \cos\theta$, but it is easier to use the smaller angle.

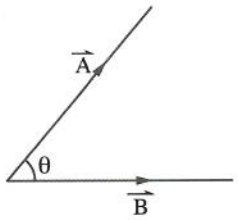
2. To identify the angle between two vectors , the two vectors must be both getting in (or out) of the common point.

For example : If the two directed segments , the end point of one of them "O" is the starting point of the other , then the smaller angle between the two vectors is the included angle between one of them and the extension of the other as shown in the following figures :

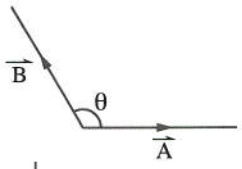


3. If θ is the angle between the two vectors $\vec{A}$ and $\vec{B}$ where $\theta \in [0 , \pi]$, then :

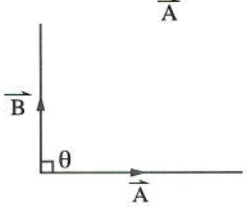
(1) $\vec{A} \cdot \vec{B} > 0$ (positive)
, if $0° < \theta < 90°$



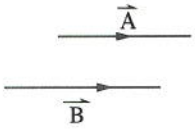
(2) $\vec{A} \cdot \vec{B} < 0$ (negative)
, if $90° < \theta < 180°$



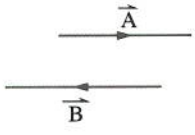
(3) $\vec{A} \cdot \vec{B} = 0$, if $\theta = 90°$
i.e. $\vec{A}$ and $\vec{B}$ are perpendicular.



(4) $\vec{A} \cdot \vec{B} = \|\vec{A}\| \|\vec{B}\|$, if $\theta = 0°$
i.e. $\vec{A}$ and $\vec{B}$ are parallel and in the same direction
and so $\vec{A} \cdot \vec{A} = \|\vec{A}\|^2$



(5) $\vec{A} \cdot \vec{B} = -\|\vec{A}\| \|\vec{B}\|$, if $\theta = 180°$
i.e. $\vec{A}$ and $\vec{B}$ are parallel but in opposite directions.



4. The unit of measuring the scalar quantity $\vec{A} \cdot \vec{B}$ equals the product of the measuring units of norm of $\vec{A}$ and $\vec{B}$

• Properties of scalar product :

1. Commutative property $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$

2. Distributive property $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$

3. If k is a real number , then $(k\,\vec{A}) \cdot \vec{B} = \vec{A} \cdot (k\,\vec{B}) = k\,(\vec{A} \cdot \vec{B})$

4. If one of the two vectors $\vec{A}$ or $\vec{B}$ or both equals zero , then $\vec{A} \cdot \vec{B} = 0$ because the norm of the zero vector = 0

i.e. $\vec{A} \cdot \vec{O} = \vec{O} \cdot \vec{A} = \vec{O} \cdot \vec{O} = 0$

5. If $\vec{A} \cdot \vec{B} = 0$, then either

• $\vec{A}$ or $\vec{B}$ or both equals zero. • $\vec{A}$ and $\vec{B}$ are perpendicular.

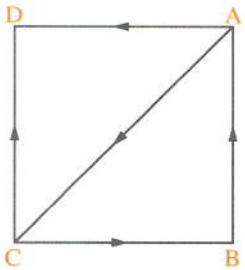
Example 1

ABCD is a square of side length 5 cm. Find :

1. $\overrightarrow{BA} \cdot \overrightarrow{CD}$ **2.** $\overrightarrow{AD} \cdot \overrightarrow{CB}$ **3.** $\overrightarrow{AD} \cdot \overrightarrow{BA}$

4. $\overrightarrow{AD} \cdot \overrightarrow{AC}$ **5.** $\overrightarrow{AC} \cdot \overrightarrow{BA}$

Solution



1. $\overrightarrow{BA}$, $\overrightarrow{CD}$ are parallel and act in the same direction

$\therefore$ θ (the included angle) = 0°

$\therefore \overrightarrow{BA} \cdot \overrightarrow{CD} = \| \overrightarrow{BA} \| \| \overrightarrow{CD} \| \cos 0° = 5 \times 5 \times 1 = 25$

2. $\overrightarrow{AD}$, $\overrightarrow{CB}$ are parallel but act in opposite directions

$\therefore$ θ (the included angle) = 180°

$\therefore \overrightarrow{AD} \cdot \overrightarrow{CB} = \| \overrightarrow{AD} \| \| \overrightarrow{CB} \| \cos 180° = 5 \times 5 \times -1 = -25$

3. $\overrightarrow{AD}$, $\overrightarrow{BA}$ are perpendicular

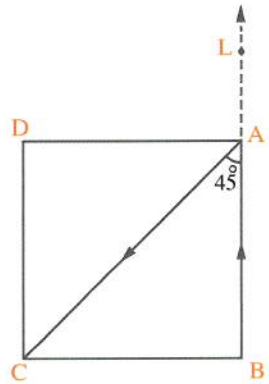
$\therefore \overrightarrow{AD} \cdot \overrightarrow{BA}$ = zero

4. $\overrightarrow{AD}$, $\overrightarrow{AC}$ includes a small angle $\angle$ DAC of measure 45° , $\| \overrightarrow{AC} \| = 5\sqrt{2}$ cm.

(diagonal of the square)

$\therefore \overrightarrow{AD} \cdot \overrightarrow{AC} = \| \overrightarrow{AD} \| \| \overrightarrow{AC} \| \cos 45° = 5 \times 5\sqrt{2} \times \frac{1}{\sqrt{2}} = 25$

5. To get the smaller angle between the two vectors $\overrightarrow{AC}$ and $\overrightarrow{BA}$, draw them getting out of (A) So $\angle$ LAC is the smaller angle between them and its measure = 135°



$\therefore \overrightarrow{AC} \cdot \overrightarrow{BA} = \| \overrightarrow{AC} \| \| \overrightarrow{BA} \| \cos 135°$

$= 5\sqrt{2} \times 5 \times -\frac{1}{\sqrt{2}} = -25$

Another solution

$\overrightarrow{AC} \cdot \overrightarrow{BA} = \overrightarrow{AC} \cdot (-\overrightarrow{AB}) = -(\overrightarrow{AC} \cdot \overrightarrow{AB})$

$= -\| \overrightarrow{AC} \| \| \overrightarrow{AB} \| \cos 45° = -5\sqrt{2} \times 5 \times \frac{1}{\sqrt{2}} = -25$

Example 2

ABC is an equilateral triangle of side length 9 cm. Find :

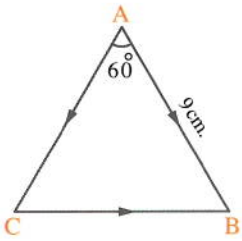
1. $\overrightarrow{AB} \cdot (\overrightarrow{AC} + \overrightarrow{CB})$ **2.** $(3\overrightarrow{BC}) \cdot (4\overrightarrow{CB})$ **3.** $(\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA}) \cdot \overrightarrow{AB}$

Solution

1. $\overrightarrow{AB} \cdot (\overrightarrow{AC} + \overrightarrow{CB}) = \overrightarrow{AB} \cdot \overrightarrow{AC} + \overrightarrow{AB} \cdot \overrightarrow{CB}$

$= \| \overrightarrow{AB} \| \| \overrightarrow{AC} \| \cos 60° + \| \overrightarrow{AB} \| \| \overrightarrow{CB} \| \cos 60°$

$= 9 \times 9 \times \frac{1}{2} + 9 \times 9 \times \frac{1}{2} = 81$



Another solution

$\because \overrightarrow{AC} + \overrightarrow{CB} = \overrightarrow{AB}$

$\therefore \overrightarrow{AB} \cdot (\overrightarrow{AC} + \overrightarrow{CB}) = \overrightarrow{AB} \cdot \overrightarrow{AB} = \| \overrightarrow{AB} \|^2 = 9^2 = 81$

2. $(3\overrightarrow{BC}) \cdot (4\overrightarrow{CB}) = 3 \times 4 (\overrightarrow{BC} \cdot \overrightarrow{CB}) = 12 \| \overrightarrow{BC} \| \| \overrightarrow{CB} \| \cos 180°$

$= 12 \times 9 \times 9 \times -1 = -972$

3. $\because \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \overrightarrow{O}$

$\therefore (\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA}) \cdot \overrightarrow{AB} = \overrightarrow{O} \cdot \overrightarrow{AB} = 0$

Example 3

If $\overrightarrow{A}$, $\overrightarrow{B}$ are two vectors , prove that :

1. $(\overrightarrow{A} - \overrightarrow{B}) \cdot (\overrightarrow{A} + \overrightarrow{B}) = \| \overrightarrow{A} \|^2 - \| \overrightarrow{B} \|^2$

2. $\| \overrightarrow{A} + \overrightarrow{B} \|^2 = \| \overrightarrow{A} \|^2 + \| \overrightarrow{B} \|^2 + 2 \| \overrightarrow{A} \| \| \overrightarrow{B} \| \cos \theta$

Solution

1. $(\vec{A} - \vec{B}) \cdot (\vec{A} + \vec{B}) = \vec{A} \cdot \vec{A} + \vec{A} \cdot \vec{B} - \vec{B} \cdot \vec{A} - \vec{B} \cdot \vec{B}$

$= \|\vec{A}\|^2 - \|\vec{B}\|^2$

2. $\|\vec{A} + \vec{B}\|^2 = (\vec{A} + \vec{B}) \cdot (\vec{A} + \vec{B}) = \vec{A} \cdot \vec{A} + \vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{A} + \vec{B} \cdot \vec{B}$

$= \|\vec{A}\|^2 + 2(\vec{A} \cdot \vec{B}) + \|\vec{B}\|^2$

$= \|\vec{A}\|^2 + \|\vec{B}\|^2 + 2\|\vec{A}\|\|\vec{B}\| \cos\theta$

- **The scalar product of the fundamental unit vectors :**

If $\hat{i}$, $\hat{j}$, $\hat{k}$ are the fundamental unit vectors of an orthogonal coordinate system , then

- $\hat{i} \cdot \hat{i} = \|\hat{i}\|\|\hat{i}\| \cos 0° = (1)(1)(1) = 1$
- $\hat{i} \cdot \hat{j} = \|\hat{i}\|\|\hat{j}\| \cos 90° = (1)(1)(0) = 0$

, in general :

- $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$
- $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{i} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{j} = \hat{i} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$

- **The scalar product of two vectors in an orthogonal coordinates system :**

If $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$,

$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$

$\therefore \vec{A} \cdot \vec{B} = (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \cdot (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$

$= A_x \hat{i} \cdot (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) + A_y \hat{j} \cdot (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) + A_z \hat{k} \cdot (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$

$= A_x B_x (\hat{i} \cdot \hat{i}) + A_x B_y (\hat{i} \cdot \hat{j}) + A_x B_z (\hat{i} \cdot \hat{k})$

$+ A_y B_x (\hat{j} \cdot \hat{i}) + A_y B_y (\hat{j} \cdot \hat{j}) + A_y B_z (\hat{j} \cdot \hat{k})$

$+ A_z B_x (\hat{k} \cdot \hat{i}) + A_z B_y (\hat{k} \cdot \hat{j}) + A_z B_z (\hat{k} \cdot \hat{k})$

$\therefore \boxed{\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z}$

Example 4

Find $\vec{A} \cdot \vec{B}$ in each of the following cases :

1. $\vec{A} = (-2, 3, 5)$, $\vec{B} = (7, 2, -1)$

2. $\vec{A} = 3\hat{i} - 4\hat{j} + \hat{k}$, $\vec{B} = -\hat{i} + 4\hat{k}$

3. $\vec{A} = (-5, 1, -2)$, $\vec{B} = 4\hat{i} + 5\hat{j}$

4. $\vec{A} = (2, -2, 1)$, $\|\vec{B}\| = 5$, $\theta = 60°$

Solution

1. $\vec{A} \cdot \vec{B} = (-2, 3, 5) \cdot (7, 2, -1) = -2 \times 7 + 3 \times 2 + 5 \times -1 = -13$

2. $\vec{A} \cdot \vec{B} = (3, -4, 1) \cdot (-1, 0, 4) = 3 \times -1 + -4 \times 0 + 1 \times 4 = 1$

3. $\vec{A} \cdot \vec{B} = (-5, 1, -2) \cdot (4, 5, 0) = -5 \times 4 + 1 \times 5 + -2 \times 0 = -15$

4. $\because \vec{A} = (2, -2, 1)$ $\quad \therefore \|\vec{A}\| = \sqrt{2^2 + (-2)^2 + 1^2} = 3$

$\therefore \vec{A} \cdot \vec{B} = \|\vec{A}\| \|\vec{B}\| \cos\theta = 3 \times 5 \times \cos 60° = 7.5$

• The angle between two vectors :

Let $\vec{A}$ and $\vec{B}$ be two non-zero vectors , including the smaller angle θ

$\therefore \vec{A} \cdot \vec{B} = \|\vec{A}\| \|\vec{B}\| \cos\theta$

$\therefore$ $\cos\theta = \dfrac{\vec{A} \cdot \vec{B}}{\|\vec{A}\| \|\vec{B}\|}$ where $0° \le \theta \le 180°$

Notice that

- $\cos\theta = 1$ then $\vec{A} \,//\, \vec{B}$ and act in the same direction
- $\cos\theta = -1$ then $\vec{A} \,//\, \vec{B}$ and act in opposite directions
- $\cos\theta = 0$ then $\vec{A} \perp \vec{B}$

Example 5

Find the measure of the included angle between the two vectors :

$\vec{A} = 3\hat{i} - 4\hat{j} + \sqrt{11}\,\hat{k}$, $\vec{B} = (-1, -2, 2)$

Solution

$\because \|\vec{A}\| = \sqrt{3^2 + (-4)^2 + \left(\sqrt{11}\right)^2} = 6$, $\|\vec{B}\| = \sqrt{(-1)^2 + (-2)^2 + 2^2} = 3$,

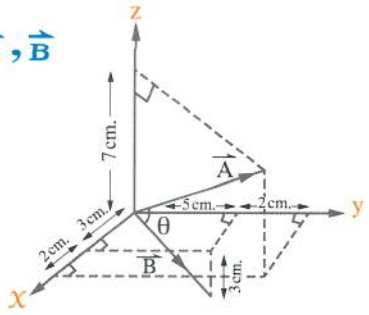
$\vec{A} \cdot \vec{B} = \left(3, -4, \sqrt{11}\right) \cdot (-1, -2, 2) = -3 + 8 + 2\sqrt{11} = 5 + 2\sqrt{11}$

$\therefore \cos\theta = \dfrac{\vec{A} \cdot \vec{B}}{\|\vec{A}\| \|\vec{B}\|} = \dfrac{5 + 2\sqrt{11}}{6 \times 3}$ $\quad \therefore \theta \simeq 49° 44' 15''$

Example 6

In the opposite figure :

Find the measure of the angle θ between the two vectors $\vec{A}$, $\vec{B}$



Solution

$\because \vec{A} = (5 , 7 , 7) , \vec{B} = (3 , 5 , -3)$

$$\therefore \cos\theta = \frac{(5 , 7 , 7) . (3 , 5 , -3)}{\sqrt{25+49+49}\ \sqrt{9+25+9}}$$

$$= \frac{15+35-21}{\sqrt{123}\sqrt{43}} = \frac{29}{\sqrt{123}\sqrt{43}}$$

$\therefore \theta \simeq 66° 29\grave{} 58\grave{}\grave{}$

Example 7

Prove that : Δ ABC is a right-angled triangle at B where

A (2 , – 1 , 3) , B (– 2 , 5 , 1) , C (– 4 , 4 , 2)

Solution

$\because \overrightarrow{BA} = \vec{A} - \vec{B} = (2 , -1 , 3) - (-2 , 5 , 1) = (4 , -6 , 2)$,

$\overrightarrow{BC} = \vec{C} - \vec{B} = (-4 , 4 , 2) - (-2 , 5 , 1) = (-2 , -1 , 1)$

$$\therefore \cos B = \frac{(4 , -6 , 2) . (-2 , -1 , 1)}{\sqrt{16+36+4}\ \sqrt{4+1+1}} = \frac{-8+6+2}{\sqrt{56}\ \sqrt{6}} = 0$$

$\therefore$ m ($\angle$ B) = 90°

$\therefore$ Δ ABC is a right-angled triangle at B

Remarks

1. You have studied before that the component of vector $\vec{A}$ in direction of vector $\vec{B}$ denoted as $A_B = \|\vec{A}\| \cos\theta$ where θ is the measure of the smaller angle included between the two vectors $\vec{A}$, $\vec{B}$

, $\because \cos\theta = \dfrac{\vec{A} . \vec{B}}{\|\vec{A}\|\|\vec{B}\|}$ $\quad \therefore A_B = \|\vec{A}\| \times \dfrac{\vec{A} . \vec{B}}{\|\vec{A}\|\|\vec{B}\|}$ $\quad \therefore A_B = \dfrac{\vec{A} . \vec{B}}{\|\vec{B}\|}$

i.e. The component of vector $\vec{A}$ in direction of vector $\vec{B}$ is $\boxed{A_B = \dfrac{\vec{A} . \vec{B}}{\|\vec{B}\|}}$

and also it is called the algebraic component of vector $\vec{A}$ in direction of vector $\vec{B}$

Or the projection of vector $\vec{A}$ in direction of vector $\vec{B}$

2. The vector component of vector $\overrightarrow{A}$ in direction of vector $\overrightarrow{B}$

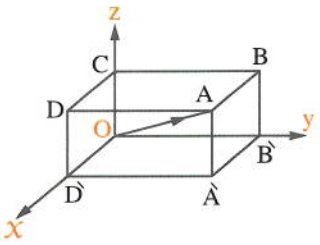
= (algebraic component of vector $\overrightarrow{A}$ in direction of vector $\overrightarrow{B}$) × (the unit vector in direction of vector $\overrightarrow{B}$)

$$= \frac{\overrightarrow{A} \cdot \overrightarrow{B}}{\| \overrightarrow{B} \|} \left(\frac{\overrightarrow{B}}{\| \overrightarrow{B} \|} \right) = \left(\frac{\overrightarrow{A} \cdot \overrightarrow{B}}{\| \overrightarrow{B} \|^2} \right) \overrightarrow{B}$$

Example 8

In the opposite figure :

ABCDÀB̀OD̀ is a cuboid , A (3 , 4 , 5) Find :



1. The component of vector $\overrightarrow{OA}$ in direction of vector $\overrightarrow{D\grave{B}}$

2. The vector component of vector $\overrightarrow{OA}$ in direction of vector $\overrightarrow{D\grave{B}}$

Solution

From the geometry of the figure

A = (3 , 4 , 5) , B = (0 , 4 , 5) , D̀ = (3 , 0 , 0)

$\therefore \overrightarrow{OA} = (3 , 4 , 5)$, $\overrightarrow{D\grave{B}} = \overrightarrow{B} - \overrightarrow{D} = (0 , 4 , 5) - (3 , 0 , 0) = (-3 , 4 , 5)$

1. The component of $\overrightarrow{OA}$ in direction of $\overrightarrow{D\grave{B}} = \dfrac{\overrightarrow{OA} \cdot \overrightarrow{D\grave{B}}}{\| \overrightarrow{D\grave{B}} \|} = \dfrac{(3 , 4 , 5) \cdot (-3 , 4 , 5)}{\sqrt{9 + 16 + 25}}$

$$= \frac{-9 + 16 + 25}{5\sqrt{2}} = \frac{32}{5\sqrt{2}} = \frac{16\sqrt{2}}{5}$$

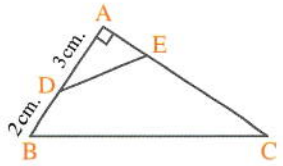
2. The vector component of vector $\overrightarrow{OA}$ in direction of vector $\overrightarrow{D\grave{B}}$

$$= \left(\frac{\overrightarrow{OA} \cdot \overrightarrow{D\grave{B}}}{\| \overrightarrow{D\grave{B}} \|^2} \right) \overrightarrow{D\grave{B}} = \frac{16}{25} (-3 , 4 , 5) = \left(\frac{-48}{25} , \frac{64}{25} , \frac{16}{5} \right)$$

Example 9

In the opposite figure :

ABC is a right-angled triangle at A and E ∈ $\overline{AC}$, D ∈ $\overline{AB}$, AD = 3 cm. , BD = 2 cm.



Find : $\overrightarrow{BA} \cdot \overrightarrow{DE}$

Solution

$$\overrightarrow{BA} \cdot \overrightarrow{DE} \equiv \overrightarrow{BA} \cdot \left(\overrightarrow{DA} + \overrightarrow{AE} \right) = \overrightarrow{BA} \cdot \overrightarrow{DA} + \overrightarrow{BA} \cdot \overrightarrow{AE}$$

$$= 5 \times 3 \cos 0 + 0 = 15$$

Notice that

We solve without getting the measure of the angle between $\overrightarrow{BA}$, $\overrightarrow{DE}$

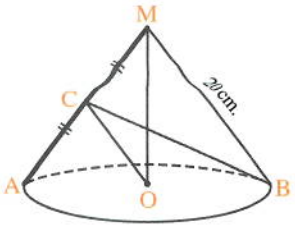
Another solution

$$\overrightarrow{BA} \cdot \overrightarrow{DE} = \| \overrightarrow{BA} \| \times \left(\text{the projection of } \overrightarrow{DE} \text{ on } \overrightarrow{BA} \right) = BA \times DA = 5 \times 3 = 15$$

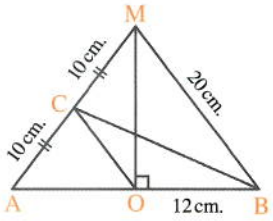
Example 10

In the opposite figure :

A right circular cone , its base circumference is 24π cm. and the length of its drawer = 20 cm , $\overline{AB}$ is a diameter of its base whose centre (O). **Find :** $\overrightarrow{BC} \cdot \overrightarrow{CO}$



Solution



$\because$ The base circumference = 24π cm.

$\therefore 2\pi r = 24\pi$

$\therefore$ r = 12 cm. and from the figure

$\therefore MO = \sqrt{(20)^2 - (12)^2} = 16$ cm.

, $\because$ O is the midpoint of $\overline{AB}$, C is the midpoint of $\overline{AM}$

$\therefore$ OC = 10 cm.

, $\cos(\angle MBO) = \frac{12}{20} = \frac{3}{5}$

$\therefore \overrightarrow{BC} \cdot \overrightarrow{CO} = (\overrightarrow{BO} + \overrightarrow{OC}) \cdot \overrightarrow{CO}$

$= \overrightarrow{BO} \cdot \overrightarrow{CO} + \overrightarrow{OC} \cdot \overrightarrow{CO}$

$= \overrightarrow{BO} \cdot \overrightarrow{CO} - \overrightarrow{OC} \cdot \overrightarrow{OC}$

$= \| \overrightarrow{BO} \| \| \overrightarrow{CO} \| \cos(\angle COB) - \| \overrightarrow{OC} \|^2 = 12 \times 10 \times \frac{-3}{5} - 10^2 = -172$

Notice that

$\cos(\angle COB) = -\cos(\angle MBO)$

$= \frac{-12}{20} = \frac{-3}{5}$

because $\overline{MB} \parallel \overline{OC}$

Another solution :

$\overrightarrow{BC} \cdot \overrightarrow{CO} = \left[\frac{1}{2}\left(\overrightarrow{BM} + \overrightarrow{BA}\right)\right] \cdot \left[-\frac{1}{2}\overrightarrow{BM}\right]$

$= -\frac{1}{4}\left[\| \overrightarrow{BM} \|^2 + \overrightarrow{BA} \cdot \overrightarrow{BM}\right]$

$= -\frac{1}{4}\left[400 + 24 \times 20 \cos B\right]$

$= -\frac{1}{4}\left[400 + 24 \times 20 \times \frac{12}{20}\right] = -172$

Third solution using the coordinates:

B (0 , 12 , 0) , O (0 , 0 , 0) , A (0 , – 12 , 0) , C (0 , – 6 , 8)

$\overrightarrow{BC} = \overrightarrow{C} - \overrightarrow{B} = (0 , -18 , 8)$, $\overrightarrow{CO} = \overrightarrow{O} - \overrightarrow{C} = (0 , 6 , -8)$

$\overrightarrow{BC} \cdot \overrightarrow{CO} = (0 , -18 , 8) \cdot (0 , 6 , -8) = -172$

